Speed, Distance & Time

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: A car travels 100 km in 2 hours. What is its speed?
Method & answer: Speed = distance ÷ time = 100 ÷ 2 = 50 km/h.

Standard: A train travels at 45 km/h for 2 hours 30 minutes. How far does it travel?
Method & answer: 2 hours 30 minutes = 2.5 hours. Distance = speed × time = 45 × 2.5 = 112.5 km.

Advanced: A plane covers 900 km in 1 hour 30 minutes. What is its average speed?
Method & answer: 1 hour 30 minutes = 1.5 hours. Speed = distance ÷ time = 900 ÷ 1.5 = 600 km/h.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A car travels 140 km in 2 hours. What is its speed?

2. A cyclist rides at 18 km/h for 2 hours. How far does she travel? A) 32 km B) 36 km C) 40 km D) 44 km

3. A train covers 250 km in 5 hours. What is its speed?

4. A car travels at 60 mph for 2 hours. How far does it travel?

5. Which of these is a sensible speed for a car on a motorway? A) 8 km/h B) 15 km/h C) 110 km/h D) 900 km/h

6. A plane flies 720 km in 1.5 hours. What is its speed?

7. A walker moves at 6 km/h for 2.5 hours. How far does she walk?

8. A bus travels 165 km in 3 hours. What is its speed?

9. Select ALL the speed calculations below that are correct: A) 120 km in 2 hours = 60 km/h B) 90 km in 1.5 hours = 60 km/h C) 150 km in 3 hours = 40 km/h D) 200 km in 2.5 hours = 80 km/h E) 63 km in 3 hours = 20 km/h

10. A car travels 210 km in 3.5 hours. What is its speed?

11. A runner covers 21 km in 3 hours. What is her speed?

12. Which speed would take the LEAST time to travel 120 km? A) 30 km/h B) 40 km/h C) 60 km/h D) 20 km/h

13. A coach travels at 65 km/h for 2 hours. How far does it travel?

14. A car takes 2.5 hours to travel 175 km. What is its average speed?

15. A cyclist travels 16 km in the first hour and 20 km in the second hour. What is her average speed over the 2 hours?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A ship sails 350 km in 5 hours. What is its speed?

6. A bus travels at 44 km/h for 2.5 hours. How far does it travel?

11. A car travels at 40 mph. Select ALL statements that are TRUE: A) In 2 hours it covers 80 miles B) In 30 minutes it covers 20 miles C) In 3 hours it covers 100 miles D) In 15 minutes it covers 10 miles E) In 1 hour 30 minutes it covers 60 miles

2. A tractor drives at 12 km/h for 3.5 hours. How far does it travel?

3. A car travels 260 km in 4 hours. What is its speed? A) 55 km/h B) 60 km/h C) 65 km/h D) 70 km/h

4. A car travels for 30 minutes at 80 km/h. How far does it travel?

5. A train travels 480 km in 4 hours. What is its speed?

7. A plane flies at 550 km/h for 2 hours. How far does it travel? A) 1,000 km B) 1,100 km C) 1,200 km D) 1,300 km

8. A cyclist covers 51 km in 3 hours. What is her speed?

9. A car travels for 15 minutes at 100 km/h. How far did it travel?

10. A walker travels at 5 km/h for 90 minutes. How far does he walk?

12. A train takes 3.5 hours to travel 315 km. What is its average speed?

13. A coach travels at 52 km/h for 2.5 hours. How far does it travel?

14. Which journey has the greatest speed? A) 180 km in 3 hours B) 140 km in 2 hours C) 252 km in 4 hours D) 99 km in 1.5 hours

15. A car travels 90 km in the first 1.5 hours and 150 km in the next 2.5 hours. What is its average speed for the whole journey?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A car travels 120 km in the first 2 hours and 100 km in the next 2 hours. What is its average speed for the whole journey?

2. A train travels 375 km at an average speed of 75 km/h. How long does the journey take? A) 4 hours B) 5 hours C) 6 hours D) 7 hours

3. A cyclist travels 24 km in 2 hours, then a further 18 km in 1 hour. What is her average speed for the whole journey?

4. A plane flies 1,050 km at a speed of 350 km/h. How long does the flight take?

5. A car travels for 2 hours at 55 km/h, then for 1.5 hours at 60

6. A coach's journey is 240 km and takes 3 hours. Later it returns the same distance but takes 1 hour longer. What is its speed on the return journey? A) 48 km/h B) 60 km/h C) 70 km/h D) 80 km/h

7. A ship sails at 30 km/h for 3 hours, then speeds up to 45 km/h for 2 hours. How far does it sail in total?

8. Two cars leave the same place at the same time, travelling in opposite directions. One travels at 55 km/h, the other at 65 km/h. How far apart are they after 3 hours?

9. A journey of 360 km is completed at an average speed of 60 km/h. Select ALL of the following that are TRUE: A) The journey takes 6 hours B) In 2 hours, 120 km is covered C) Doubling the speed to 120 km/h would halve the time to 3 hours D) At 60 km/h, half the journey (180 km) takes 4 hours E) In 4 hours, 240 km is covered

10. A car's journey is 420 km. For the first half it travels at 70 km/h

11. A cyclist travels 45 km at 15 km/h, rests for 1 hour, then travels a further 40 km at 20 km/h. How long does the whole trip take, including the rest?

12. A train leaves at 09:00 travelling at 84 km/h. How far has it travelled by 11:45? A) 210 km B) 231 km C) 252 km D) 273 km

13. A car travels 320 km in 4 hours for the first part of a journey, then a further 160 km in 2 hours. What is its average speed for the whole journey?

14. A cyclist and a runner both set off from the same point at the same time, travelling in the same direction. The cyclist rides at 28 km/h and the runner jogs at 10 km/h. How far apart are they after 2.5 hours?

15. A lorry travels 480 km. It covers the first 280 km at 70 km/h. At what

km/h. How far does it travel in total?

and for the second half at 60 km/h. How long does the whole journey take?

speed must it travel the remaining 200 km to complete the whole journey in exactly 8 hours?